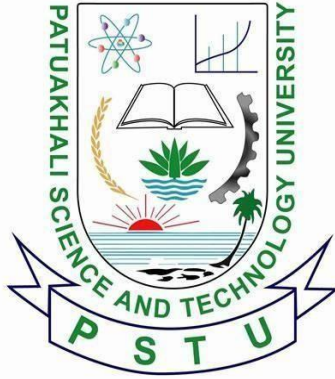




PATUAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY

CCE 310 :: Software Development Project-I Sessional Project Proposal

Project Title: PetBondhuBD

Submission Date: Fri 18, July 2025

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PetBondhuBD

1. Abstract

PetBondhuBD is a **cross-platform Flutter app** addressing the fragmented pet care ecosystem in Bangladesh. It integrates **health tracking, AI-powered lost pet recovery, and an essentials marketplace** into one intuitive platform. Targeting 5M+ pet owners, the app leverages Firebase and Google Vision AI to deliver localized solutions for vaccination reminders, emergency alerts, and doorstep delivery of pet supplies.

2. Objectives

Primary: Develop a unified pet care app for Android/iOS with:

-  **Multi-pet profiles** (breed, age, medical history)
-  **AI lost pet finder** (photo matching + geo-alerts)
-  **E-commerce integration** (food, meds, accessories)

Key Metrics:

- Reduce vaccine missed doses by 70%
- Achieve 80% lost pet recovery rate via AI
- Onboard 100+ Dhaka-based pet stores in Phase 1

3. Problem Statement

Bangladeshi pet owners face:

-  **No centralized platform:** 3+ apps needed for basic care.
-  **Inefficient lost pet recovery:** Reliance on social media.
-  **Supply gaps:** 60% struggle to find specialized pet products locally (*Source: Dhaka Pet Owners Survey 2023*).

4. Related Work

App	Focus	Limitations
<i>PetDesk</i>	Health tracking	US-centric, no marketplace
<i>Pawscout</i>	Lost pets	No AI/BD localization
PetBondhuBD	All-in-one	Tailored for Bangladesh

5. Scope

In-Scope:

- Flutter-based mobile app (Android/iOS)
- Firebase backend (Auth, Firestore)
- Google Vision AI integration
- Dhaka-focused pilot

Out-of-Scope:

- IoT wearables (Phase 2)
- Non-Bengali language support

6. Methodology

6.1 Technology Stack

Layer	Tools
Frontend	Flutter 3.13, Dart
Backend	Firebase Auth, Firestore, Cloud Functions

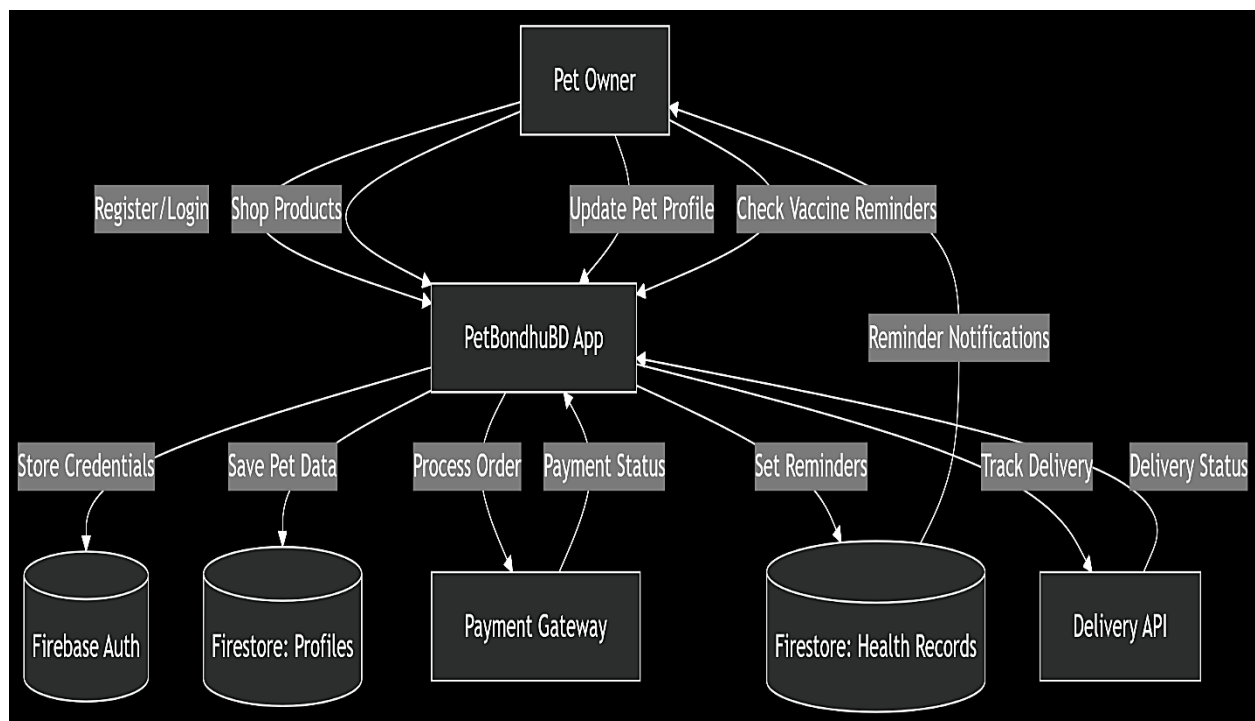
Layer	Tools
AI	Google Vision API
DevOps	GitHub Actions, Play Store/App Store

6.2 Design Principles

- **Offline-first:** Low-bandwidth functionality
- **MUIS:** Minimalist UI with Bengali typography
- **GDPR Compliance:** Encrypted pet health data

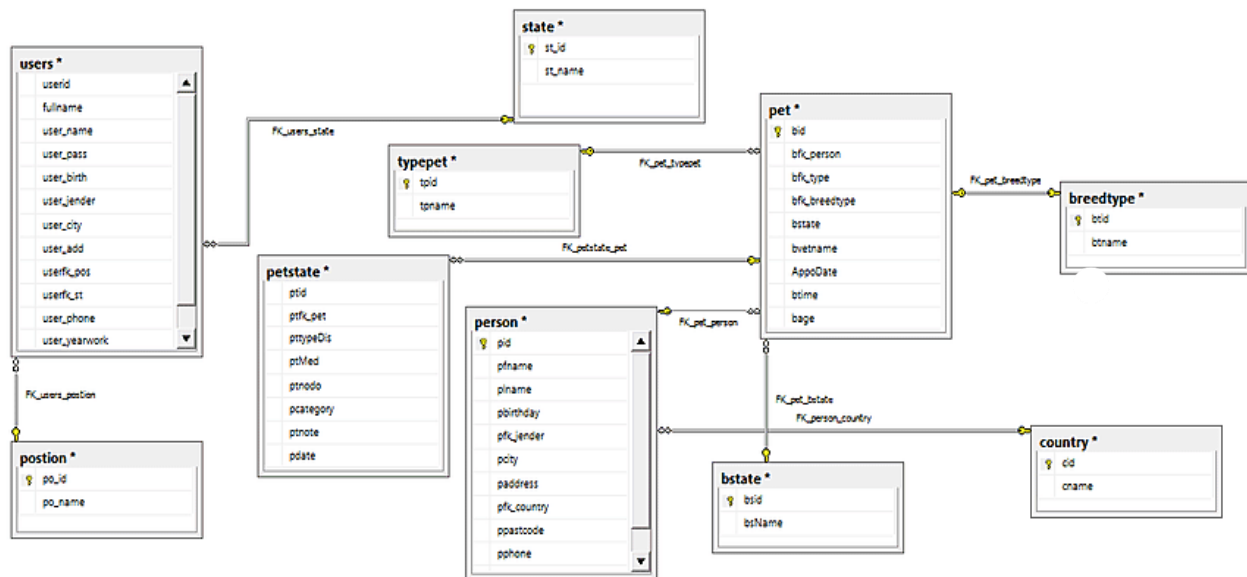
7. Visual Models

7.1 Flow Chart Diagram



7.2 ER Diagram

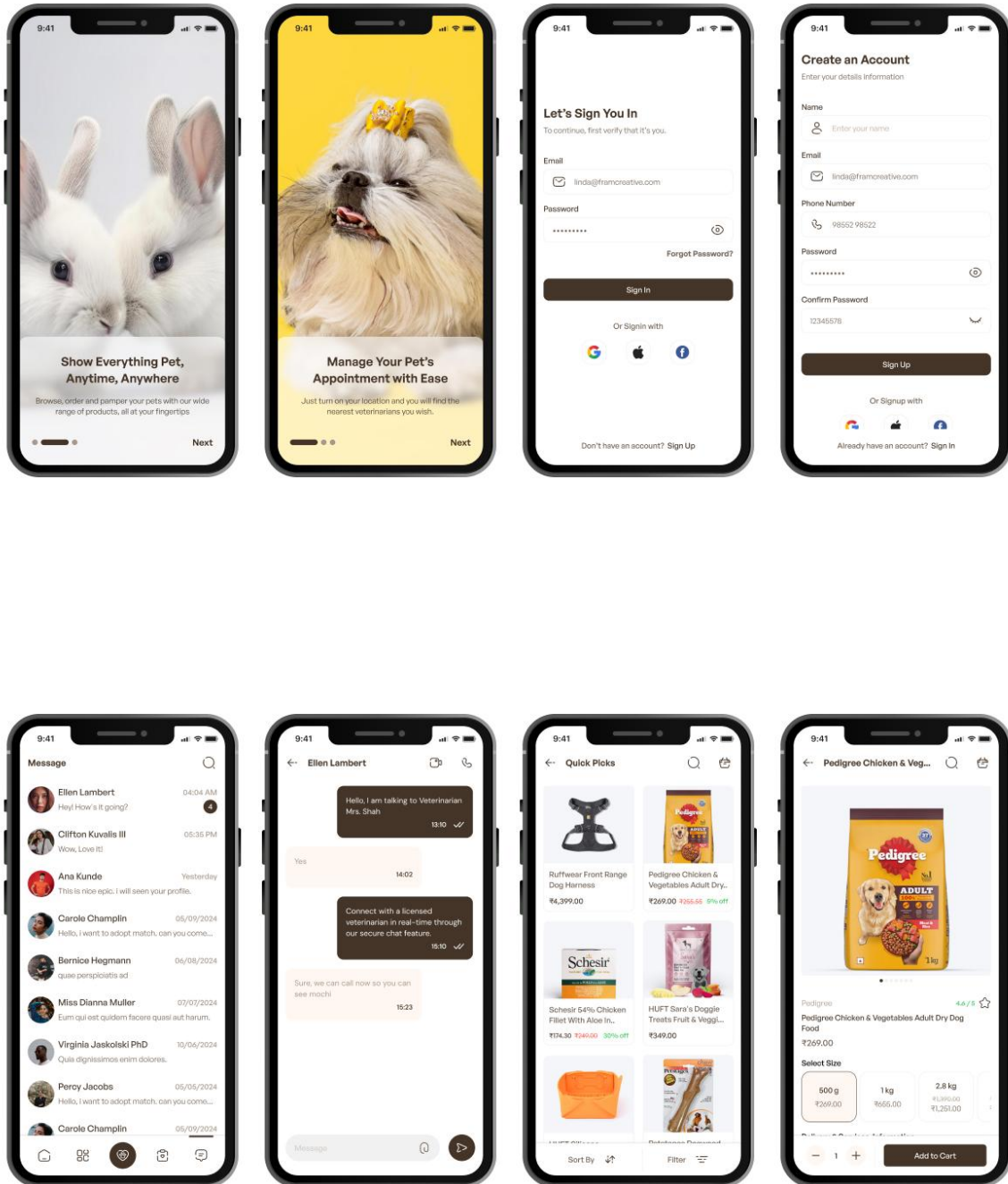
- Entities: User, Pet, Order, VaccineRecord

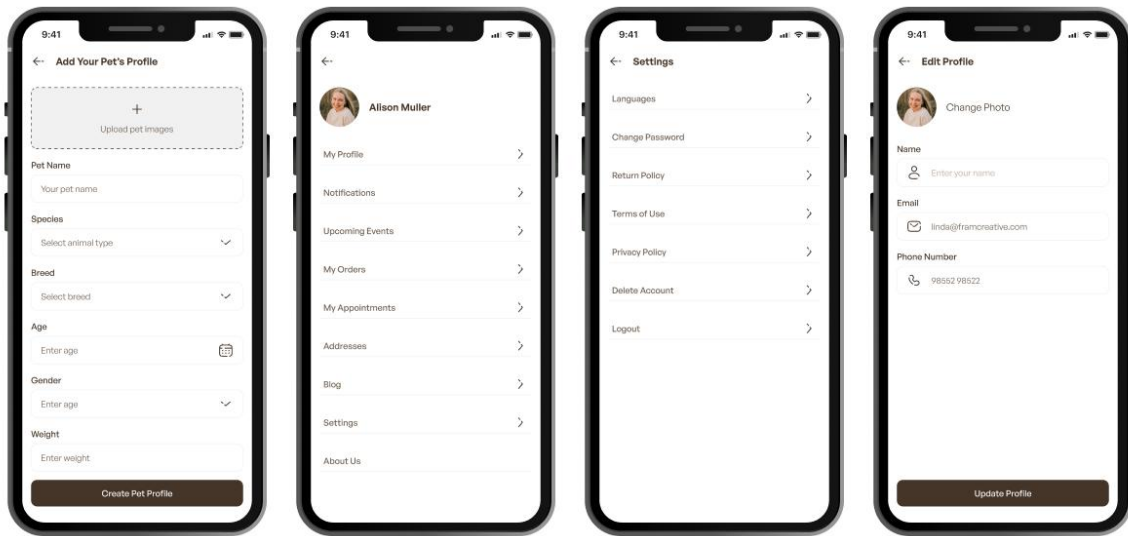


7.3 Gantt Chart

Task	Start Day	End Day	Duration	Timeline (Gantt View)
Requirements Gathering	Day 1	Day 14	2 weeks	
UI Mockups	Day 15	Day 28	2 weeks	
Firebase Auth	Day 15	Day 21	1 week	
Pet Profiles	Day 22	Day 28	1 week	
Vaccine Reminders	Day 29	Day 35	1 week	
AI Lost Pet Finder	Day 36	Day 49	2 weeks	
E-Commerce Module	Day 50	Day 63	2 weeks	
Delivery Tracking	Day 64	Day 70	1 week	

7.4 UI Mockups





8. Limitations

- 🐾 **Mixed-breed accuracy:** AI works best with purebred pets.
- 📶 **Rural connectivity:** Offline mode required.

9. Expected Results

- **10K+ downloads in 6 months**
- **40% fewer lost pets (AI matching)**
- **70% more vaccines (auto-reminders)**
- **Impact:** Saves time, builds pet community.